

Feature Engineering and Feature Selection Validation Report

Model Name

Mortgage Insurance Credit Risk Model

Validation Objective

The purpose of this review is to assess whether the feature engineering and feature selection framework is methodologically sound, business-justified, reproducible, and free from leakage, bias, or unnecessary model instability.

The review focused on:

- Engineered feature construction
- Variable transformations
- Ratio and interaction features
- Missing-value related features
- Outlier handling
- Economic feature engineering
- Borrower aggregation features
- Feature leakage risks
- Feature stability and interpretability

The assessment was performed using the implemented preprocessing and feature engineering framework.

1. Feature Creation and Transformation

Objective

To assess whether engineered features are economically meaningful, appropriately transformed, and suitable for mortgage credit risk modeling.

Description

The review evaluated the logic used to construct derived variables, ratios, borrower-level metrics, and standardized variables. The assessment also considered whether transformations preserve the underlying economic meaning of the original variables.

Validation Assessment

Review Item	Status
All engineered features are documented	Partial
Business rationale for engineered features is reasonable	Yes
Transformations preserve economic meaning	Yes
Ratios and derived variables are appropriate	Yes
Derived features align with intended use	Yes
No material distortion of risk relationship identified	Yes

Validator Comments

The framework creates several engineered features relevant to mortgage credit risk, including:

- Loan-to-income ratio (LTI)
- Monthly interest rate
- Monthly payment calculation
- Property age
- Property value
- Rural classification
- Economic-region assignment

The engineered variables are economically meaningful and generally consistent with mortgage underwriting practices.

Additional documentation of feature definitions and business rationale would improve governance transparency.

2. Missing-Value Related Features

Objective

To assess whether missing-value handling and related feature construction are appropriate and consistent.

Description

The review evaluated whether missing-value treatment could affect engineered feature behavior or introduce bias into the model development process. The assessment also considered whether missingness itself may contain predictive information.

Validation Assessment

Review Item	Status
Missing-value flags considered where appropriate	Partial
Missingness assessed as potentially informative	Partial
Imputation logic documented and justified	Partial
Imputation avoids future information	Yes
Missing-value treatment consistent across datasets	Yes

Validator Comments

The framework uses several fallback methods to reduce missing values, including:

- Bureau score substitution
- Group-level propagation
- Formula-based reconstruction
- Group-average replacement

Credit scores are derived from multiple bureau sources using fallback logic. Missing property values are reconstructed using LTV and insurance information.

No evidence of future-information usage was identified in the imputation logic. However, formal documentation of missing-value methodology should be improved.

3. Outlier-Related Features and Handling

Objective

To assess whether outlier treatment is methodologically reasonable and does not distort engineered feature relationships.

Description

The review evaluated how extreme observations are handled during feature engineering and whether outlier treatment may unintentionally suppress valid tail-risk behavior relevant to mortgage credit risk.

Validation Assessment

Review Item	Status
Extreme values reviewed for business validity	Yes
Capping and truncation documented	Partial
Tail-risk observations preserved appropriately	Partial
Outlier indicator variables justified	Partial
Outlier treatment consistent across datasets	Yes

Validator Comments

The framework applies range clipping and hard limits for:

- Credit scores
- Property values
- Contract rates
- Loan terms

Credit scores outside acceptable ranges are replaced with missing values, while contract rates are normalized to realistic levels.

The validator considers the overall approach reasonable; however, selected thresholds should be formally documented and justified.

4. Categorical Variable Handling

Objective

To assess whether categorical variables are grouped, standardized, and encoded appropriately.

Description

The review assessed how categorical variables are normalized, grouped, and transformed for use in the modeling framework. The assessment also considered whether category handling may introduce instability or leakage.

Validation Assessment

Review Item	Status
Category grouping documented	Partial
Rare categories handled appropriately	Partial
Ordinal and nominal variables treated appropriately	Yes
Encoding method suitable for model type	Yes
Encoding avoids leakage	Yes
Category definitions stable over time	Partial

Validator Comments

The framework standardizes several categorical variables, including:

- Loan purpose
- Portability indicators
- Occupancy indicators
- Tenure mapping
- Regional assignment

Unknown categories are replaced using standardized labels. Regional segmentation and rural classification are applied consistently.

Additional documentation of category grouping logic would improve governance transparency.

5. Time-Based Feature Engineering

Objective

To assess whether time-based features are constructed appropriately and free from future-information leakage.

Description

The review evaluated the creation of date-based, lagged, and economic features to ensure that only information available at the underwriting date is used in feature generation.

Validation Assessment

Review Item	Status
Date variables converted appropriately	Yes
Lag features use prior information only	Partial
Rolling features constructed appropriately	Partial
Vintage effects considered	Yes
No future information used in time-based features	Partial

Validator Comments

The framework creates time-based variables using:

- RECEIVED_SID
- BOOKYEAR
- Economic lag variables
- Economic forward-shift variables

Economic variables include lagged and shifted HPI and unemployment measures.

Potential leakage risk exists if forward-looking economic variables are not aligned correctly with the application date. Additional validation testing is recommended.

6. Interaction and Ratio Features

Objective

To assess whether interaction and ratio features are mathematically correct, economically meaningful, and appropriate for mortgage credit risk modeling.

Description

The review evaluated affordability ratios, borrower-level combinations, and interaction features used to capture borrower risk characteristics and economic relationships.

Validation Assessment

Review Item	Status
Interaction terms are meaningful	Yes
Ratios are mathematically correct	Yes
Multicollinearity assessed where applicable	Partial
Interaction features improve interpretability	Yes
Number of interaction features reasonable	Yes

Validator Comments

The framework creates several economically meaningful ratio features, including:

- Loan-to-income ratio
- Monthly payment calculations
- Property-value reconstruction
- Weighted bureau score combinations

The engineered features are aligned with mortgage underwriting and borrower affordability assessment.

No excessive feature complexity was identified during review.

7. Binning and Discretization

Objective

To assess whether variable grouping and discretization methods are stable, interpretable, and appropriately justified.

Description

The review evaluated whether grouped variables and discretized categories are based on reasonable business or statistical logic and whether the resulting bins remain interpretable over time.

Validation Assessment

Review Item	Status
Bin definitions documented	Partial
Bins based on business or statistical logic	Yes
Bins are interpretable	Yes
Bin boundaries reasonable	Yes
Binning consistent across datasets	Yes

Validator Comments

The framework primarily relies on normalization and clipping rather than aggressive statistical binning. Several variables are grouped into standardized business categories such as:

- Rural versus urban
- Tenure type
- Loan purpose

The grouping logic appears reasonable and operationally interpretable.

8. Feature Leakage Assessment

Objective

To assess whether engineered or selected features contain direct or indirect future information that could artificially improve model performance.

Description

The review evaluated whether feature construction respected the underwriting decision date and whether any engineered variables unintentionally included post-outcome information or target proxies.

Validation Assessment

Review Item	Status
Candidate features reviewed for leakage	Partial
No post-outcome information identified	Yes
No proxy outcome variables identified	Partial
Feature construction respects as-of date	Partial
Leakage checks performed on engineered features	Partial

Validator Comments

No obvious target leakage was identified within the feature engineering framework.

However, the implementation includes forward-looking economic shift variables that may introduce leakage risk if timing alignment is not controlled properly.

Additional formal leakage testing is recommended before production use.

9. Feature Redundancy and Correlation

Objective

To assess whether redundant or highly correlated variables may affect model stability or interpretability.

Description

The review evaluated whether overlapping variables or excessive feature complexity could reduce model robustness or create instability during model development.

Validation Assessment

Review Item	Status
Highly correlated variables identified	Partial

Review Item	Status
Redundant features reviewed	Partial
Correlation impact assessed	Partial
Overlapping information rationalized	Partial
Feature complexity appropriate	Yes

Validator Comments

The framework combines multiple bureau variables and borrower-level statistics. While the resulting feature set appears manageable, no formal redundancy or correlation analysis was identified during review.

Additional feature correlation analysis is recommended prior to final model development.

10. Feature Selection Methodology

Objective

To assess whether feature selection methods are appropriate, explainable, and aligned with the intended model purpose.

Description

The review evaluated whether the selected features are supported by business rationale, statistical relevance, and operational interpretability.

Validation Assessment

Review Item	Status
Feature selection criteria documented	Partial
Selection method appropriate	Yes
Domain knowledge considered	Yes
Statistical selection methods applied consistently	Partial
Final feature set defensible	Yes

Review Item	Status
Excluded features documented	No

Validator Comments

The selected features appear primarily driven by mortgage underwriting knowledge and business relevance.

The feature set includes:

- Credit bureau variables
- Affordability measures
- Property characteristics
- Regional economic variables
- Borrower aggregation metrics

The validator considers the final feature set generally appropriate for mortgage insurance credit risk modeling. However, formal documentation of feature-selection methodology would strengthen governance.

11. Stability and Robustness of Features

Objective

To assess whether engineered and selected features remain stable across time periods, segments, and changing portfolio conditions.

Description

The review evaluated whether feature distributions and relationships remain reasonably stable over different vintages and market environments.

Validation Assessment

Review Item	Status
Feature distributions assessed over time	Partial
Stability reviewed by segment and vintage	Partial

Review Item	Status
Features stable across sample periods	Partial
Features robust to small data changes	Yes
Stability concerns documented	Partial

Validator Comments

The framework includes multiple years of data and regional segmentation, which supports future stability analysis.

However, no formal feature stability monitoring or population stability testing was identified during review. Additional monitoring procedures are recommended.

12. Interpretability and Governance

Objective

To assess whether the engineered and selected features are explainable, governable, and suitable for ongoing monitoring.

Description

The review evaluated whether the final feature set can be understood by business, governance, and validation stakeholders and whether the feature framework supports long-term monitoring and revalidation.

Validation Assessment

Review Item	Status
Final feature set explainable to stakeholders	Yes
Features comply with model risk standards	Partial
Features appropriate for model materiality	Yes
Expert overrides documented	Partial
Features support monitoring and revalidation	Yes

Validator Comments

Most engineered features are economically interpretable and aligned with mortgage underwriting practices. Examples include:

- Borrower affordability ratios
- Property characteristics
- Regional economic indicators
- Credit bureau metrics

The overall feature framework appears appropriate for governance review and ongoing model monitoring.

13. Validation Conclusion

Validation Assessment

Conclusion Item	Status
Feature engineering is acceptable	Yes
Feature engineering acceptable subject to remediation	Yes
Feature engineering is not acceptable	No

Conclusion

The feature engineering and feature selection framework is generally appropriate for mortgage insurance credit risk modeling.

Key strengths include:

- Economically meaningful feature construction
- Strong borrower aggregation logic
- Integration of macroeconomic variables
- Structured preprocessing framework
- Reproducible feature-generation process

The following improvements are recommended:

- Improve feature documentation

- Add formal leakage testing
- Perform feature correlation analysis
- Document feature-selection methodology
- Add feature stability monitoring

Overall, the feature engineering framework is considered fit for use subject to the remediation and governance improvements identified during validation.

Key Findings

- Feature engineering logic is generally consistent with mortgage underwriting practices.
- No major transformation concerns were identified.
- Economic variables improve risk sensitivity but may create leakage risk if timing is not controlled properly.
- Formal feature-selection documentation should be improved.
- Additional feature stability and correlation analysis is recommended.

Required Actions

Action	Priority	Owner
Document feature-selection methodology	Moderate	Model Development
Add formal leakage testing	High	Validation Team
Perform feature correlation analysis	Moderate	Model Development
Document feature transformation rationale	Moderate	Model Governance
Implement feature stability monitoring	Moderate	Model Risk
Validate timing of forward economic variables	High	Model Development

Suggested Validation Wording

“The feature engineering and feature selection process was reviewed to assess whether transformed and selected variables are methodologically sound, business-justified, reproducible, and free from leakage or undue bias. The validator confirmed that the final feature set is generally appropriate for the model’s intended use, subject to remediation of identified issues relating to stability, documentation, and leakage controls.”